Rohan Chandra | TS/SCI Clearance

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Summary

Robotics software and ML engineer with two years of production ML experience and hands-on ROS 2 work across mobile robots, drones, and manipulators. Builds LLM-driven robot agents, real-time embedded pipelines, and the ML infrastructure behind them. Completing a Johns Hopkins M.S. in Robotics and Autonomous Systems (Dec 2026).

Education

Johns Hopkins University, M.S. Robotics and Autonomous Systems | GPA 3.96, Sep 2023 – Expected Dec 2026
- Coursework: Embodied Agentic AI for Robotic Control, Kinematics & Dynamics of Robots, Software Development for Real-Time Embedded Systems, Deep Learning with PyTorch, Reinforcement Learning
- Teaching Assistant, Introduction to Robotics (2 semesters): PID control, EKF, SLAM, sampling-based planning

University of Virginia, B.S. Computer Engineering, High Distinction, Minor in Business | GPA 3.67, Aug 2018 – May 2022

Experience

AI/Machine Learning Engineer, Bana Solutions, Tysons, VA, Oct 2024 – Present
- Won 1st place in company-wide hackathon with an original VLM / visual question answering concept that lets analysts search thousands of images for specific information; awarded three SPOT performance bonuses
- Optimized a graph-based Autoencoder using Apache Spark and PyArrow, reducing model training time 20x
- Designed and deployed a Job Role Clustering model using graph embeddings and transformer attention mechanisms
- Orchestrated ML pipelines with Apache Airflow; tracked model artifacts and deployment readiness with MLflow; monitored data and model drift with Evidently

Associate Software Engineer, Northrop Grumman, Dulles, VA, Sep 2022 – Oct 2024
- Delivered algorithm improvements in Python and C++ to a proprietary tool supporting classified satellite programs
- Developed multi-threaded applications and 15+ Dockerized microservices; improved Jenkins CI/CD pipeline performance and maintained unit test coverage

ML/Robotics Research Assistant, Prof. Alemzadeh, University of Virginia, Jan 2021 – May 2022
- Built an image augmentation pipeline that more than doubled a surgical robotics training dataset; integrated it into Mask R-CNN to improve semi-autonomous robotic surgery model performance

FIRST Robotics Software Coach, Nova Labs, Fairfax, VA, Jul 2022 – Jan 2026
- Built the robotics program across 5 FTC teams (multiple state championship appearances); founded Robot-Go, an 8-week bootcamp with a YouTube curriculum

Projects

WALL-E: LLM-Driven Inspection Rover | Embodied Agentic AI, JHU | Fall 2026
- Built a natural-language-controlled differential-drive robot: an Ollama-hosted LLM runs a ReAct reasoning loop through ROSA and ROS 2, calling custom tools that wrap the robot's sensors and actuators
- Implemented PID motor control and micro-ROS firmware on a Raspberry Pi Pico 2W, MCP servers for tool access, and real-time object detection on the onboard camera feed
- Co-authored a paper defining a controlled evaluation protocol against external ground truth: encoder-targeted vs. timed stopping, wheel-only vs encoder-IMU heading fusion, and ultrasonic vs. monocular ranging

Real-Time Drone Telemetry Pipeline | Real-Time Embedded Systems, JHU | Spring 2026
- Streamed roll/pitch/yaw (BNO055 IMU over I2C), GPS, and live video from a Pixhawk 6C / Raspberry Pi 5 quadcopter to a ground-station dashboard over TCP using length-prefixed frames and JSON telemetry

Independent Quadcopter Platform (Holybro X500 V2) Spring 2026
- Building a PX4 quadcopter with a Pixhawk 6C linked over UART/MAVLink to a Raspberry Pi 5 companion (ROS 2 Humble, Hailo AI HAT+); configured PX4 failsafes and arming; developed ROS 2 offboard control in PX4 SITL/Gazebo

Deep Learning Steganography | Deep Learning Developments with PyTorch, JHU | Summer 2025
- Co-developed a 5.3M-parameter multi-branch autoencoder trained in L*a*b space with a perceptual ΔE2000 loss; hides a full image at 17.6 bpp (cover PSNR 43.6 dB) with chance-level detectability (AUC ≈ 0.49)

RAG-Powered AI Agent | GSA Federal AI Hackathon, Top-10 Finalist
- Built a RAG agent with Cohere embeddings and auto-generated source bibliographies to make scattered USDA.gov articles queryable in natural language; presented to ~30 stakeholders and press

Skills

- **Robotics:** ROS 2, ROS, MoveIt, Gazebo, RViz, PX4, MAVLink, QGroundControl, micro-ROS, ROSA, SLAM, EKF, PID
- **ML & AI:** PyTorch, TensorFlow, OpenCV, CNNs, Transformers, VLMs, LLM agents (Ollama, MCP), RL, RAG, Mask R-CNN
- **MLOps & Infrastructure:** Airflow, MLflow, Evidently, Spark, Docker, Kubernetes, Jenkins, Linux, Git
- **Hardware:** Raspberry Pi 5, Pixhawk 6C, ESP32, Arduino, Hailo AI HAT+, IMU/GPS/cameras, FreeRTOS, UART/I2C
- **Languages:** Python, C++ | Spoken: English (native), Japanese, Hindi, Spanish